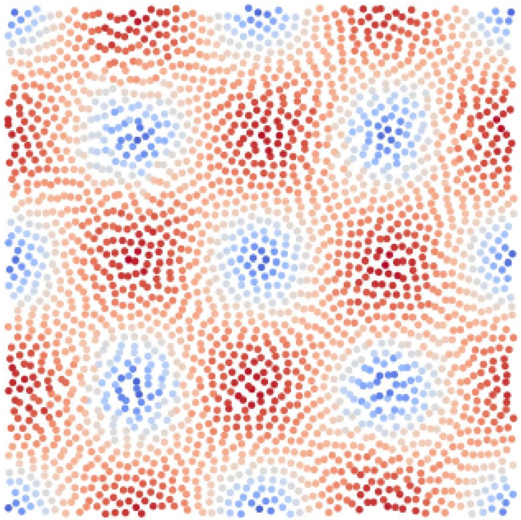
Correlation-Aware Training for Lagrangian Fluid Emulators with Continuous Convolutions

Guided Research Project

Yusuf Miraç Uyar
29.10.2025

Overall Structure

- Background Information
- Symmetric Fourier Basis Convolution
- Lagrangebench TGV–2D
- Dataset and Setup
- Experiments and Results
- Conclusion and Future Work
- Questions



Background Information

Smoothed Particle Hydrodynamics (SPH)

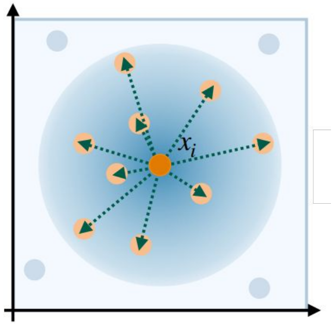
$$\rho_i = \sum_j m_j W(\|\mathbf{x}_i - \mathbf{x}_j\|, h)$$

- SPH represents a continuum by particles of mass m_i at positions x_i
- Kernel-weighted neighbor sums over a compact support of radius h

Background Information

Graph Neural Networks (GNN)

- Model fluids in a Lagrangian view where particles are graph nodes and edges connect neighbors within a cutoff radius
- Edge features are built from relative geometry and node features carry fluid attributes

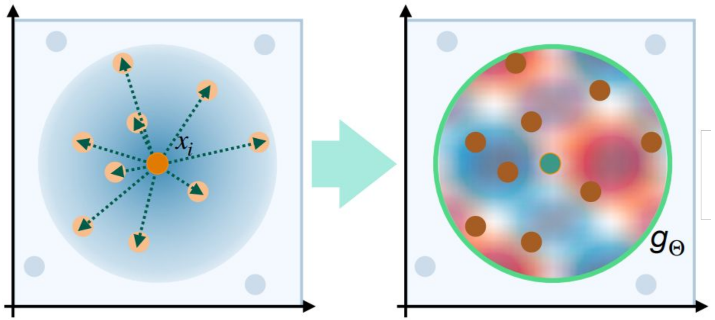


single message passing step for particle i

Background Information

Continuous Convolutions

- A subset of GNNs and utilize only coordinate distances as inputs to the filter functions
- Combines those filters with the features of adjacent vertices to form the messages



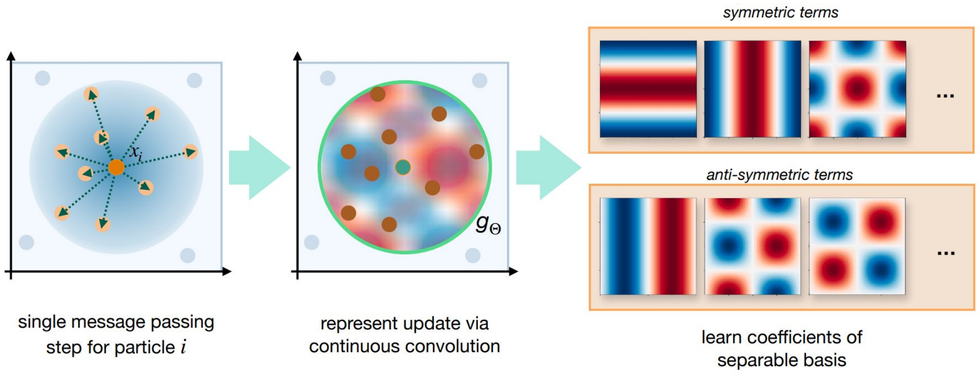
single message passing step for particle i

represent update via continuous convolution

Symmetric Fourier Basis Convolution (SFBC)

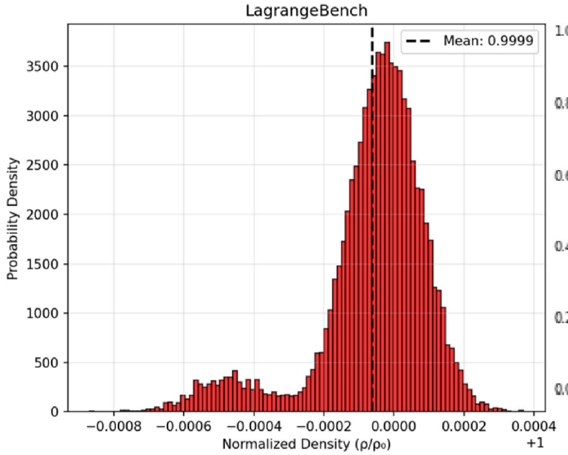
$$\mathcal{L}_{\text{SFBC}} = \frac{1}{N}\sum_{i=1}^{N} \|\hat{\mathbf{z}}_i - \mathbf{z}_i\|_2^2$$

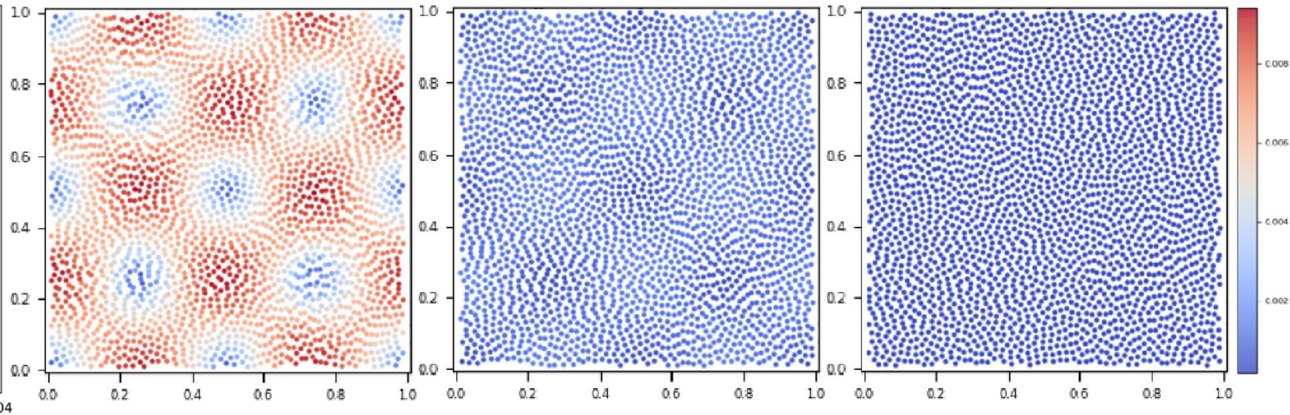
- Formulates particle interactions via continuous, symmetry-aware basis convolutions.
- Achieves robust accuracy on SPH-style tasks
- Loss objective is a per-sample mean-squared error (MSE)



Lagrangebench TGV–2D

- We use TGV-2D dataset from Lagrangebench paper.
- A canonical vortical flow used to probe coherence and decay behavior
- SPH-generated trajectories with coarse-grained prediction task (every 100 steps)





Taylor Green vortex velocity snapshot at step = 10, 60, and 113

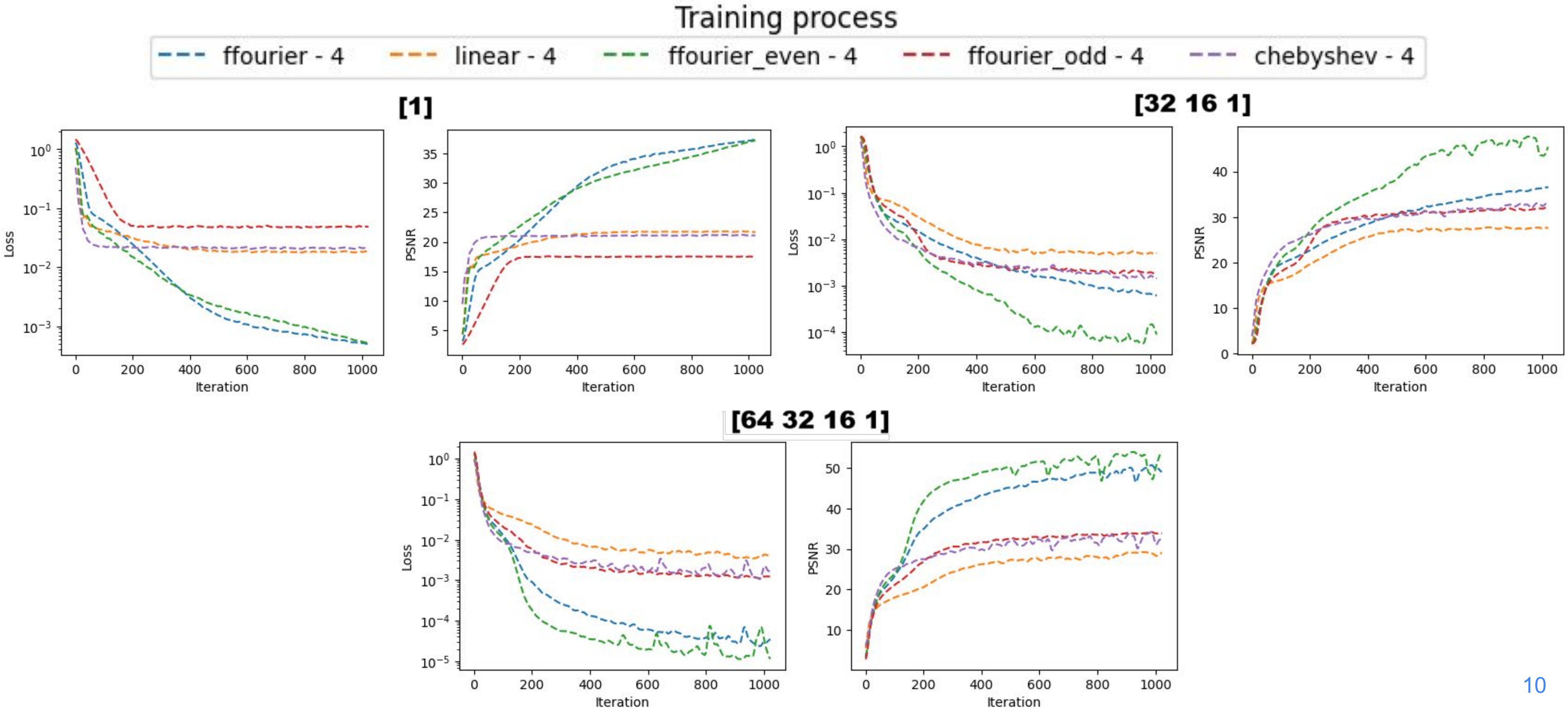
Dataset and Setup

- 2500 particles evolving on $[0, 1]^2$
- 100 training and 50 test trajectories
- Each trajectory has 126 frames
- Experimented on (11 variants in total, from [1] upto [64, 32, 16, 4, 1] with 1–5 message-passing layers
- AdamW optimizer with decaying learning rate ranging between: [0.01, 0.0001]

Experiments and Results (MSE Loss)

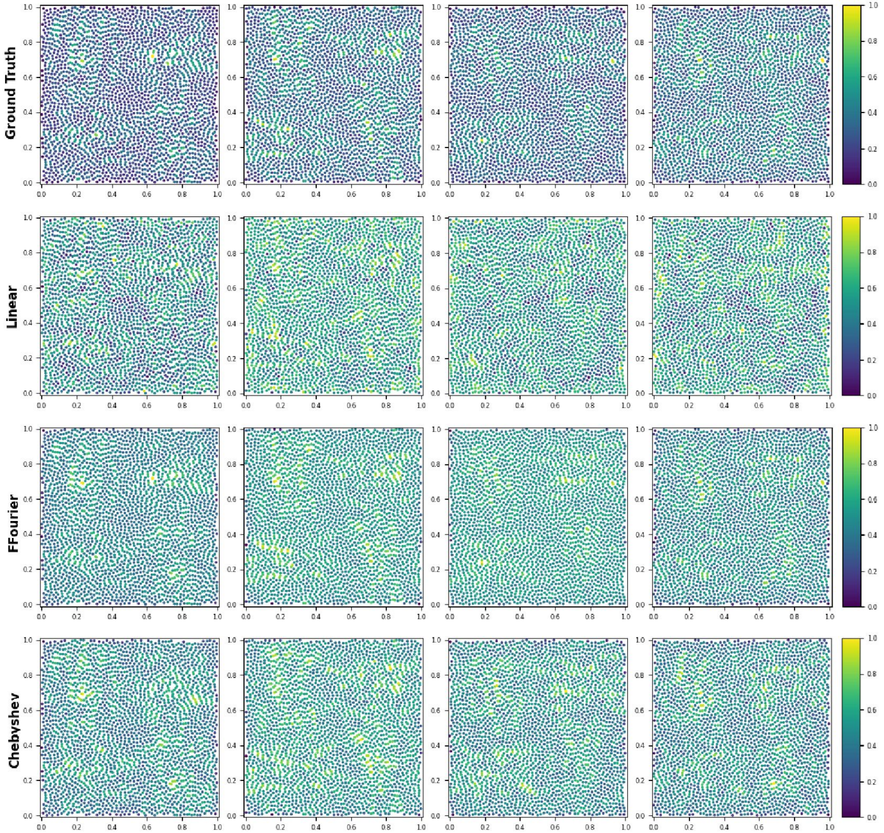
- Adding depth consistently reduces loss and improves PSNR, with marked gains by 3–4 layers
- The lowest MSE in our sweep appears at 4 layers for Fourier models
- Fourier models converge faster and to significantly lower loss than linear/Chebyshev
- Losses as absolute values are hard to interpret across dataset
- Lower MSE does not always yield better density prediction visuals.

Experiments and Results (MSE Convergence)



Experiments and Results (MSE Loss)

Architecture	Basis	MSE	PSNR (dB)
[1]	linear	2.17e-02	21.45
[1]	chebyshev	2.58e-02	20.69
[1]	fourier	**5.65e-04**	**37.31**
[32,16,1]	linear	6.89e-03	26.45
[32,16,1]	chebyshev	2.17e-03	31.59
[32,16,1]	fourier	**8.66e-04**	**35.60**
[64,32,16,1]	linear	5.57e-03	27.58
[64,32,16,1]	chebyshev	1.94e-03	31.94
[64,32,16,1]	fourier	**3.19e-05**	**49.82**



[32 16 1] architecture prediction on pure MSE objective

Correlation Loss

We used Pearson correlation as a metric:

$$\operatorname{corr}(\hat{\boldsymbol{\phi}}, \boldsymbol{\phi}) = \frac{\sum_{i=1}^{M} \left(\hat{\phi}_i - \overline{\hat{\phi}}\right)\left(\phi_i - \overline{\phi}\right)}{\sqrt{\sum_{i=1}^{M} \left(\hat{\phi}_i - \overline{\hat{\phi}}\right)^2} \sqrt{\sum_{i=1}^{M} \left(\phi_i - \overline{\phi}\right)^2}}$$

Reformed correlation as a loss term:

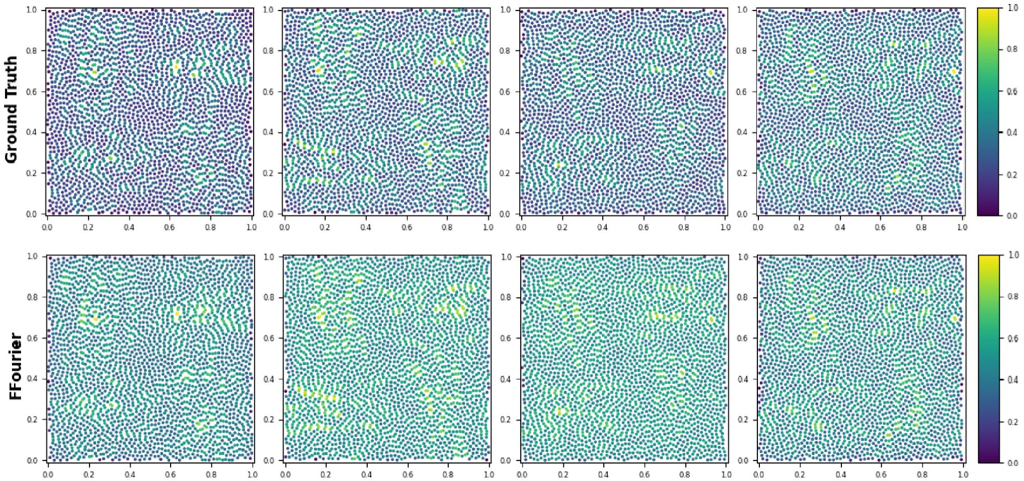
$$\mathcal{L}_{\text{corr}} = 1 - \frac{1}{B} \sum_{b=1}^{B} \operatorname{corr}\left(\hat{\rho}^{(b)}, \rho^{(b)}\right)$$

And, another loss term to make it scale-invariant:

$$\mathcal{L}_{\text{scale}} = \frac{1}{B} \sum_{b=1}^{B} \left(\frac{\overline{\hat{\rho}}^{(b)} - \overline{\rho}^{(b)}}{\overline{\rho}^{(b)} + \varepsilon} \right)^2$$

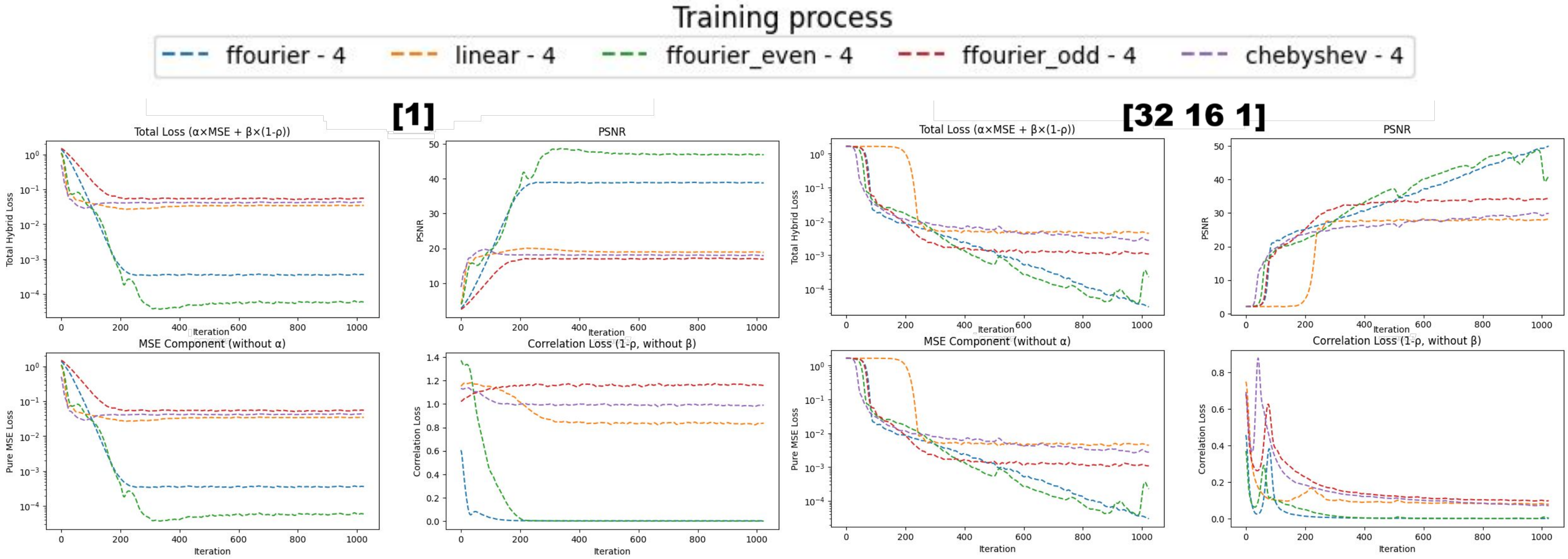
We introduced Hybrid Loss as:

$$\mathcal{L}_{\text{hyb}} = \alpha \, \mathcal{L}_{\text{SFBC}} + \beta \, \mathcal{L}_{\text{corr}} + \lambda \, \mathcal{L}_{\text{scale}}$$



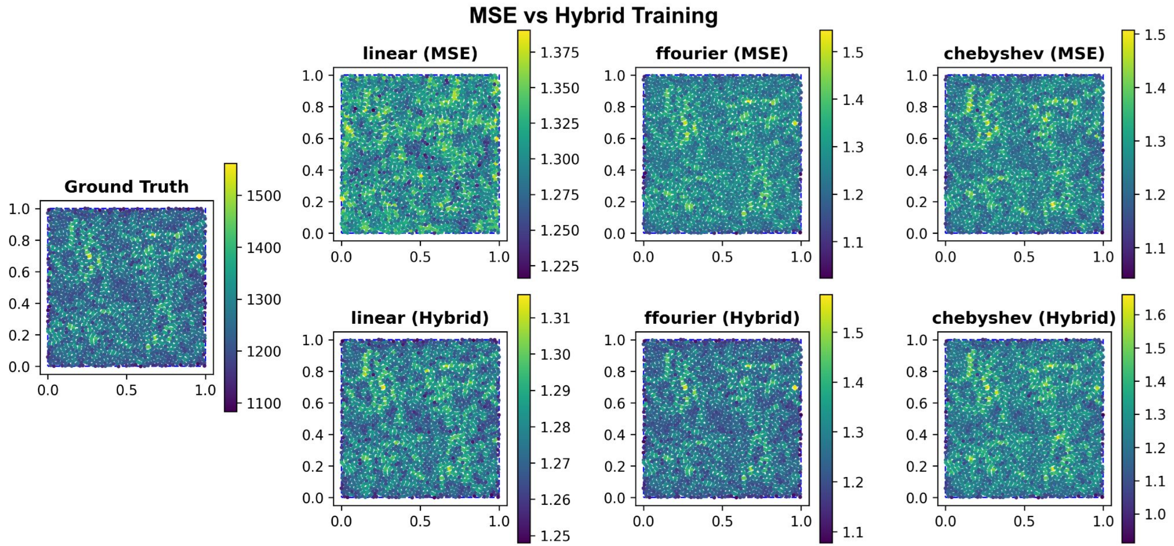
[32 16 1] architecture prediction on pure MSE objective

Experiments and Results (Hybrid Loss Convergence)



Experiments and Results (Hybrid Loss)

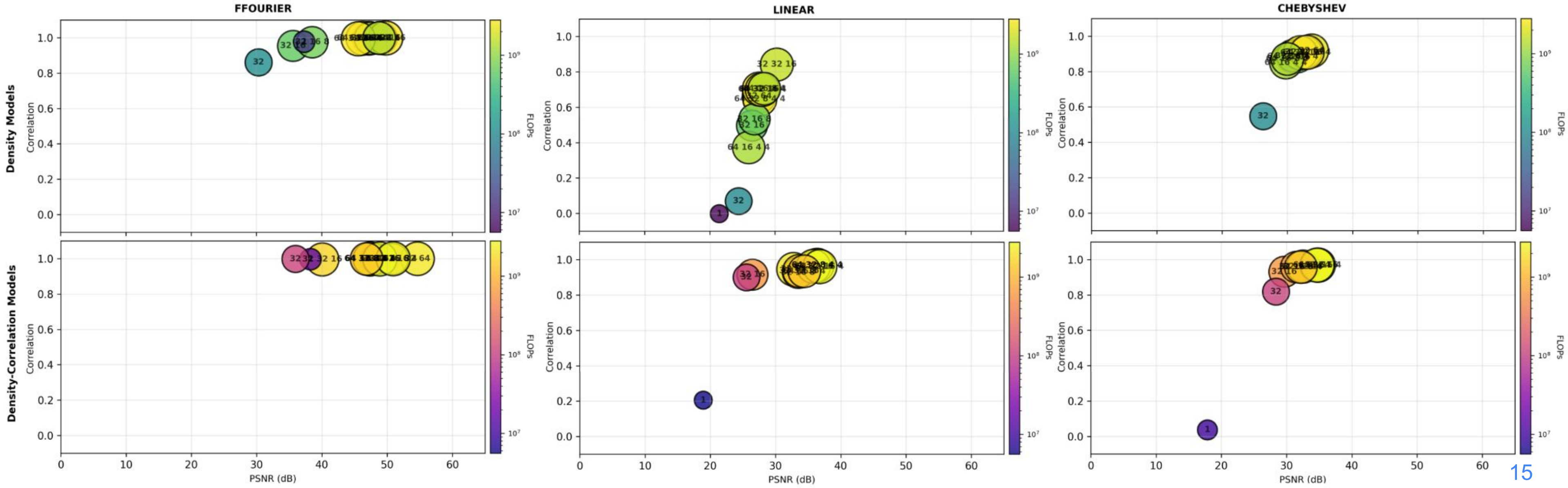
- Better correlation translates to better visuals
- Hybrid objective consistently increases correlation and often reduces MSE
- Strongest for Fourier models at shallow–mid depth



MSE vs Hybrid Training on [32 16 1] Fourier architecture

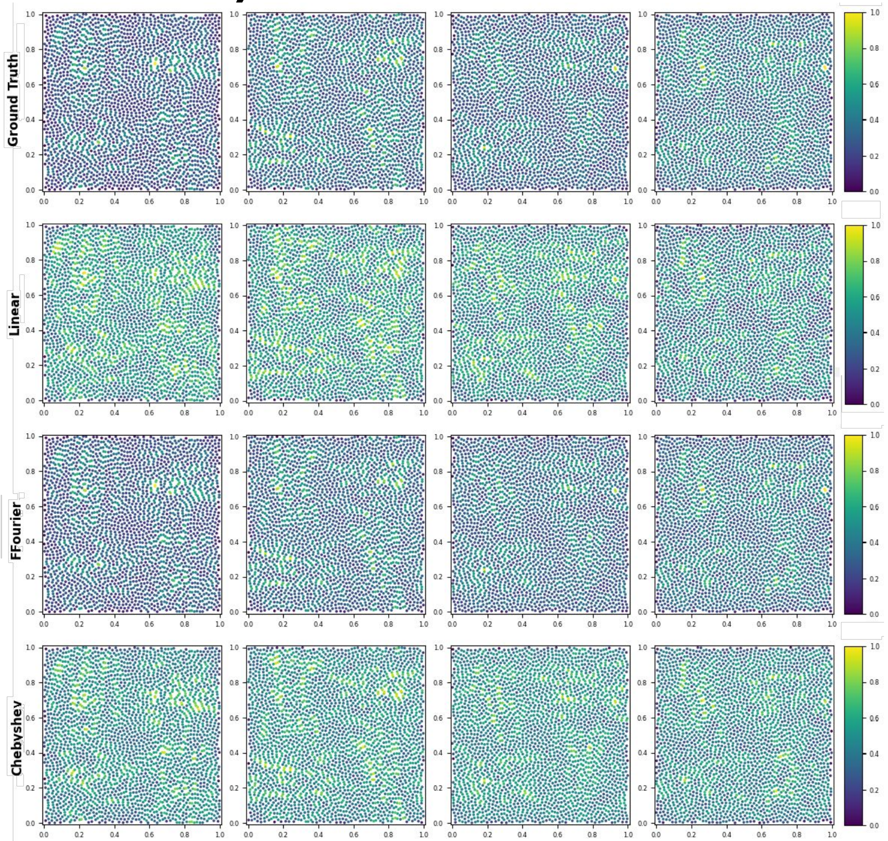
Experiments and Results (Hybrid Loss)

- Hybrid objective consistently increases correlation and often reduces MSE
- Strongest for Fourier models at shallow–mid depth
- More stable and robust training



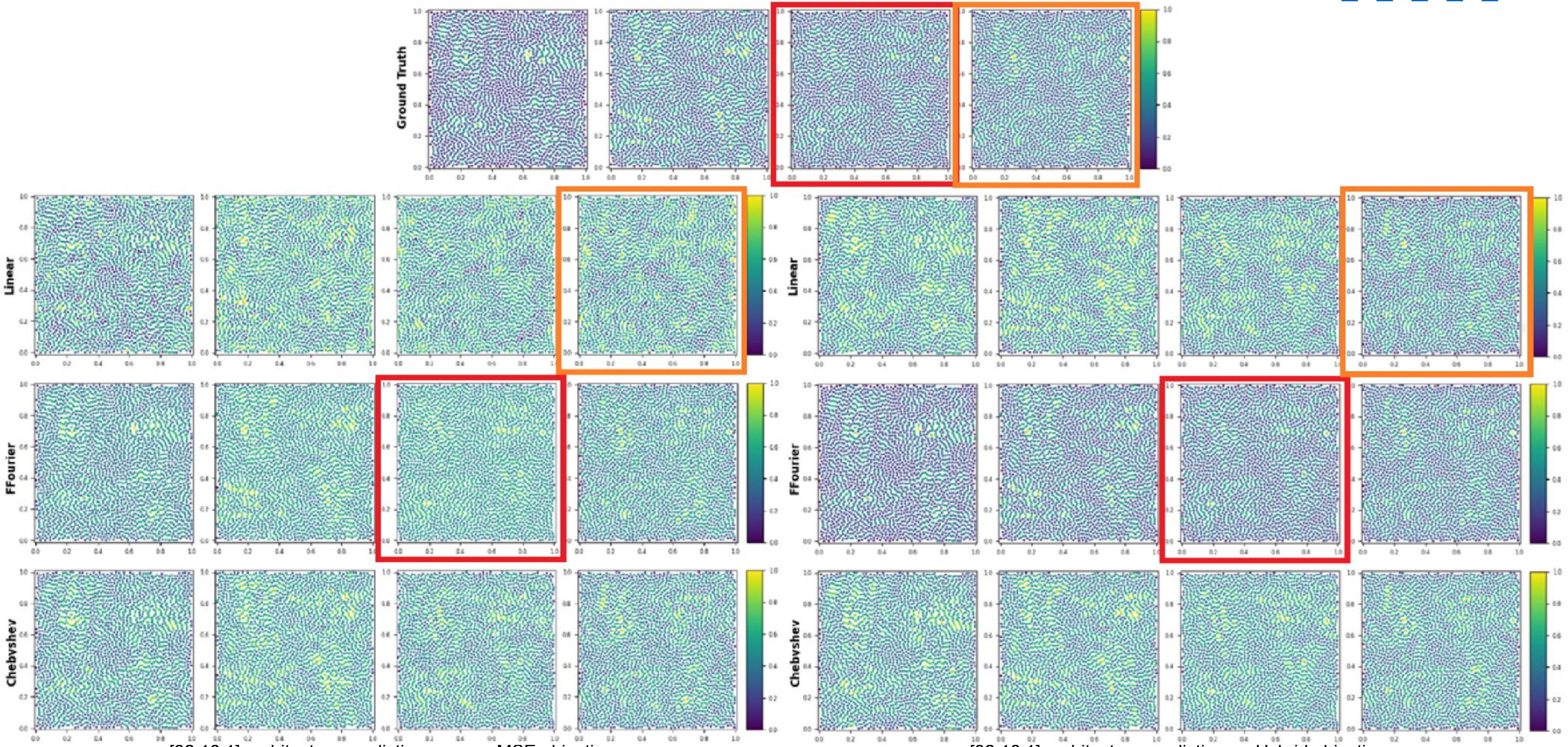
Experiments and Results (Hybrid Loss)

Architecture	MSE — Corr	MSE — Corr (Hybrid)	FLOPs (M)
Linear			
linear [1]	2.17×10^{-2} — -0.002	3.87×10^{-2} — 0.206	6
linear [32, 16, 1]	6.89×10^{-3} — 0.498	7.12×10^{-3} — 0.916	601
linear [32, 16, 8, 1]	6.43×10^{-3} — 0.533	1.48×10^{-3} — 0.927	723
linear [64, 32, 16, 1]	$\mathbf{5.57\times10^{-3}}$ — $\mathbf{0.704}$	$\mathbf{8.24\times10^{-4}}$ — $\mathbf{0.962}$	2718
Chebyshev			
chebyshev [1]	2.58×10^{-2} — -0.198	4.95×10^{-2} — 0.037	10
chebyshev [32, 16, 1]	2.17×10^{-3} — 0.882	3.27×10^{-3} — 0.931	609
chebyshev [32, 16, 8, 1]	2.54×10^{-3} — 0.887	2.15×10^{-3} — 0.958	733
chebyshev [64, 32, 16, 1]	$\mathbf{1.94\times10^{-3}}$ — $\mathbf{0.906}$	$\mathbf{9.84\times10^{-4}}$ — $\mathbf{0.969}$	2728
Fourier			
fourier [1]	5.65×10^{-4} — 0.978	4.53×10^{-4} — 0.998	16
fourier [32, 16, 1]	8.66×10^{-4} — 0.955	4.82×10^{-5} — $\mathbf{0.999}$	621
fourier [32, 16, 8, 1]	4.39×10^{-4} — 0.976	$\boxed{\mathbf{2.65\times10^{-5}} — \mathbf{0.999}}$	748
fourier [64, 32, 16, 1]	$\boxed{\mathbf{3.19\times10^{-5}} — \mathbf{0.998}}$	4.01×10^{-5} — $\mathbf{0.999}$	2743



[32 16 1] architecture prediction on Hybrid objective

Experiments and Results (Hybrid Loss)



[32 16 1] architecture prediction on pure MSE objective

[32 16 1] architecture prediction on Hybrid objective

Conclusion and Future Work

- Receptive field and network depth matter
- MSE alone is not enough for visual quality
- Hybrid loss increases correlation while decreasing MSE
- Hybrid loss provides more stable and robust training
- Testing on different datasets
- Extending the problem space (3D Flows)

Questions

?